

Explorative Analysis for Predicting Mobile Addiction Among Engineering Students

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Abstract

In this research, we've tried to forecast mobile addiction among engineering students in India by examining a dataset of 302 students gathered via a questionnaire. The dataset encompasses 62 attributes, including demographic data, mobile usage behaviors, and self-assessed addiction levels. Findings reveal that 48.3% of students exhibit a high risk of addiction, scoring 5 or above on a self-reported scale, with average daily mobile usage rising interestingly. Current support systems, such as counseling, are insufficient, with approximately 70% of at-risk students lacking access to adequate intervention programs. We applied machine learning methods, including K-Nearest Neighbor (KNN), Naive Bayes, Decision Tree, and Random Forest, to predict addiction levels, with Random Forest achieving the highest performance at 89% accuracy and a weighted precision of 0.891. The results underscore the pressing need for enhanced counseling, awareness initiatives, and stricter mobile usage policies in educational settings. We recommend embedding digital wellness education in curricula and leveraging technology for early intervention to address mobile addiction effectively.

Keywords: Mobile Addiction, Engineering Students, Machine Learning, Digital Wellness.

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1 Introduction

The rising incidence of mobile addiction among students in India has emerged as a significant public health concern, fueled by the widespread adoption of smartphones. This issue aligns with Sustainable Development Goal 3.4, which emphasizes mental health and well-being. Among engineering students, increased mobile usage is frequently linked to academic stress, social disconnection, and mental health challenges. Statistics indicate that approximately 60% of Indian students spend over 4 hours daily on their mobile devices.

Here, we take a closer look at a dataset of 302 engineering students from various colleges, collected through a questionnaire. The dataset includes variables such as age, gender, daily mobile usage, app preferences, and self-reported addiction scores. Notably, 48.3% of students scored 5 or higher on a 1-10 addiction scale, indicating a significant risk of mobile addiction. This trend is attributed to factors such as academic pressure, social media dependency, and the use of mobile devices as a stress-coping mechanism. Financially, excessive mobile usage impacts students, accounting for roughly 3% of their monthly expenses.

Counseling programs in Karnataka’s educational institutions are limited, with each counselor handling approximately 200 students annually, leaving 70% of high-risk students without adequate support. Although some institutions enforce “no-phone” zones during classes, inconsistent policies have failed to curb rising addiction rates. This research employs machine learning techniques to identify at-risk students and propose actionable interventions to improve support systems and foster digital well-being.

2 Related Works

2.1 Review of Literature

Gladius et al. (2017) explored the consequences of frequent mobile phone use among medical students in Tamil Nadu, focusing particularly on its impact on sleep, stress, and academic outcomes. Their findings indicated a notable connection between prolonged phone usage and reduced sleep quality, which in turn correlated with elevated stress and lower academic performance. The authors emphasized the need for structured mobile usage regulations in medical institutions to improve student well-being (1).

Haque et al. (2024) conducted a cross-sectional study analyzing how smartphone habits affect the mental health and sleep quality of Indian medical students. Their research highlighted a significant association between increased

screen time and symptoms of depression, anxiety, and insomnia. The study proposed that preventive education and awareness programs targeting digital behavior should be incorporated into medical curricula (2).

Siddharthan et al. (2024) investigated the influence of social media dependency on sleep among medical students in Tamil Nadu. The study found that excessive social media engagement was strongly linked with disturbed sleep cycles and daytime fatigue. It recommended introducing behavioral interventions and digital detox practices to support healthier online habits (3).

Krishnakumar et al. (2024) examined the psychological effects of heavy smartphone usage among healthcare students in Chennai. Their results indicated a strong link between high screen time and elevated psychological distress, including anxiety and disrupted sleep routines. The researchers advised integrating digital hygiene workshops and stress management programs within academic settings (4).

Correa-Iriarte et al. (2023) presented an international perspective by analyzing the indirect effects of smartphone use on sleep through bedtime procrastination. Their mediation model demonstrated that problematic phone use does not just harm sleep directly—it promotes behaviors like delaying bedtime, which in turn degrades sleep quality. This finding underscores the role of behavioral psychology in managing smartphone overuse (5).

Soni et al. (2017) conducted a cross-sectional study assessing smartphone addiction among adolescents and its association with sleep quality and behavioral problems. The study found that excessive smartphone use was significantly linked to poor sleep quality and increased levels of depression, anxiety, and stress among high school students (6).

Kumar et al. (2023) performed a scoping review analyzing the correlation between smartphone addiction and sleep problems in collegiate students. The review concluded that increased screen time and smartphone addiction were consistently associated with poor sleep quality, insomnia, and reduced academic performance (7).

Kurugodiyavar et al. (2017) investigated the impact of smartphone use on sleep quality among medical students in Karnataka. The study revealed that students with higher smartphone usage had significantly poorer sleep quality, showing the need for awareness programs on responsible smartphone use (8).

Balekar and Amulani (2019) explored the repercussions of smartphone addiction on social lives and mental health. Their findings indicated that excessive smartphone use led to social isolation, increased stress levels, and a decline in mental well-being among college students (9).

Kulkarni et al. (2019) examined the relationship between smartphone use,

sleep disturbances, depression, and anxiety in adolescents. The study concluded that high smartphone usage was associated with significant sleep disturbances and elevated levels of depression and anxiety, highlighting the need for interventions targeting smartphone use among adolescents (10).

3 Methodology

Let's walk through the methodology, implemented using Jupyter Notebook with Python 3.12.4.

3.1 Proposed System Model

Data collection, preprocessing, exploratory data analysis (EDA), model development, and evaluation are the steps that shape the system model. After gathering 302 engineering students' questionnaire answers, we preprocessed the data to guarantee its quality. EDA was applied to visualize addiction trends and usage patterns, and appropriate features were chosen for predictive modeling. In order to estimate the degree of mobile addiction, predictive models such as KNN, Naive Bayes, Decision Tree, and Random Forest were created.

3.2 Data Source and Preprocessing

We collected the dataset through a questionnaire from 302 engineering students, consisting of 62 attributes such as age, gender, daily mobile usage, app preferences, notification frequency, and self-reported addiction scores. Preprocessing steps:

- **Time Conversion:** Converted daily mobile usage from HH:MM format to minutes using a custom function.
- **Range Conversion:** Transformed screen time and notification ranges to midpoints (e.g., "1-2 hours" to 1.5 hours).
- **Missing Values:** Filled numerical columns with median values and categorical columns with mode values; total missing values were 2.1%.
- **Cleaning:** Validated ordinal and binary columns against predefined categories, replacing invalid entries with the mode.
- **Encoding:** Used LabelEncoder for categorical variables (e.g., gender, app preferences).

- **Scaling:** Standardized numerical features using StandardScaler for uniformity.

The preprocessed dataset was applied for both visualization and modeling.

3.3 Exploratory Data Analysis and Interpretation

EDA was performed to understand mobile usage patterns and addiction trends. We present our key findings through visualizations to provide deeper insights into the dataset.

Figure 1 below shows the distribution of self-reported addiction scores among the 302 students, with a threshold line at 5 to highlight the proportion of students at high risk.

Figure 1: Distribution of Self-Reported Addiction Scores

Inference on Self-Reported Addiction Histogram: Figure 1 gives a clear picture of how addiction scores are spread out among students. You can see that 48.3% of them scored 5 or higher, which is marked by the red line—it's almost half the group! It's a bit worrying to see so many students rating themselves this high, and it really makes you think about how big a problem mobile addiction might be on campus.

Next, we examine the prevalence of addiction by gender. Figure 2 presents a bar chart comparing the percentage of addicted students between males and females.

Figure 2: Addiction Prevalence by Gender

Inference on Gender Addiction Bar Chart: Figure 2 shows that 52% of male students are addicted compared to 43% of female students. That gap is pretty noticeable—males seem to be struggling more with mobile addiction. It makes me wonder if there's something about how guys use their phones, maybe gaming or social media habits, that's driving this difference.

To explore the relationship between mobile usage and addiction, we analyzed the correlation between daily usage time and self-reported addiction scores. Figure 3 displays a scatter plot with a regression line, highlighting a correlation of 0.65.

Figure 3: Daily Usage Time vs. Addiction Score

Inference on Usage vs. Addiction Scatter Plot: Figure 3 really brings the correlation to life—with a score of 0.65, you can see a clear upward trend. The more hours students spend on their phones, the higher their addiction scores tend to be. Some students using their phones for 8 or 10 hours a day are scoring way up at 7 or 8 on the addiction scale. It's like the phone just keeps pulling them in deeper the longer they're on it, which feels all too relatable.

Finally, we grouped the dataset by gender and residential status to compare average daily usage times. Figure 4 shows the results, revealing that hostelites report higher usage (4.2 hours) than localites (3.8 hours).

Figure 4: Average Daily Usage by Gender and Residential Status

Inference on Usage by Gender and Residence Bar Chart: Figure 4 tells an interesting story—hostelites are using their phones more, averaging 4.2 hours a day compared to 3.8 hours for localites, and this pattern holds for both males and females. It makes sense when you think about it: hostelites might be away from home, maybe feeling a bit isolated, so they turn to their phones more for entertainment or to stay connected. That extra half hour or so could really add up over time and push them closer to dependency.

4 Results and Discussion

We chose features by looking at how closely they tied to mobile addiction, picking ones that stood out, like daily phone usage time, notification frequency, app usage, and habits such as “I often check my phone for updates.” We scaled the data, split it into 90% for training and 10% for testing to keep things fair, and applied SMOTE to even out the training set. Then, we applied a method called RFE with Random Forest to narrow it down to the top 30 features.

We tested four models—KNN, Naive Bayes, Decision Tree, and Random Forest—fine-tuning them to get the best results. Random Forest came out on top, scoring 89% accuracy and 0.891 precision on the test set, with daily usage time and notification frequency showing up as the biggest factors.

4.1 Model Performance

Four machine learning models were trained and evaluated: KNN, Naive Bayes, Decision Tree, and Random Forest. Hyperparameter tuning was performed using GridSearchCV, and performance was assessed using weighted precision and accuracy.

Table 1: Model Performance Metrics

Model	Accuracy (%)	Weighted Precision	CV Precision (Mean $\pm 2 \times$ SD)
KNN	82	0.824	0.810 ± 0.032
Naive Bayes	78	0.790	0.775 ± 0.045
Decision Tree	85	0.857	0.842 ± 0.038
Random Forest	89	0.891	0.880 ± 0.025

Random Forest outperformed other models, achieving 89% accuracy and a weighted precision of 0.891. The cross-validation precision score of 0.880 indicates robust performance. Feature importance analysis revealed that daily mobile usage time, notification frequency, and the urge to check apps repeatedly were the top predictors of addiction.

4.2 Comparison with Other Models

KNN and Naive Bayes showed lower performance due to their sensitivity to feature scaling and class imbalance, respectively. Looking at Figure 5, it's clear that Random Forest really stands out as the strongest model for predicting mobile addiction among students. With an accuracy of 89% and a weighted precision of 0.891, it's noticeably ahead of the others, like KNN at 82% accuracy and Naive Bayes at 78%.

Figure 5: Comparison of Model Performance

4.3 Prediction of Mobile Addiction

The Random Forest model predicts that 48.3% of the students are at a risk of mobile addiction, consistent with the self-reported scores. Among hostelites, the addiction rate is higher (52%) compared to localites (45%), likely due to increased screen time and social isolation. The model identified key risk factors, such as spending over 4 hours daily on mobile devices and receiving more than 50 notifications per day.

4.4 Analysis

To deepen our understanding of mobile addiction patterns, we conducted a detailed analysis of key variables and their relationship with addiction levels.

Figure 6 below shows the distribution of average daily mobile usage time across all students, measured in hours. This provides insight into the extent of mobile engagement within our sample.

Figure 6: Distribution of Average Daily Mobile Usage Time

Inference on Daily Usage Histogram: Figure 6 paints a striking picture of mobile usage among students. A large chunk of them are clocking over 6 hours a day, with some even hitting up to 10 hours or more. This aligns with our finding that 48.3% of students scored 5 or higher on the addiction scale, showing that excessive usage is pretty common. It makes you think about how much time we're all spending on our phones and how that might be pulling us away from other important stuff like studying or just chilling offline.

Next, we explore the relationship between notification frequency and addiction levels. Figure 7 presents a box plot of the average daily notifications received by students classified as addicted versus those who are not.

Figure 7: Notification Frequency by Addiction Level

Inference on Notification Box Plot: Figure 7 really hits home how notifications play into addiction. Students labeled as addicted are getting way more notifications—some are even dealing with hundreds a day—compared to those who aren't addicted. The spread is wider for the addicted group too, showing that some are just overwhelmed by constant pings. It's like their phones are always buzzing, pulling them back in, which matches our model's finding that notification frequency is a top risk factor. This makes me wonder how much we're letting those little alerts control our day.

5 Conclusion

Mobile addiction among engineering students in Karnataka poses a significant challenge to mental health and academic performance. Our analysis of 302 students revealed that 48.3% are at high risk, with average daily usage exceeding 6-7 hours. The current support infrastructure, primarily counseling services, is inadequate, with 70% of high-risk students lacking access to interventions. Using machine learning, our model managed to hit 89% accuracy in predicting mobile addiction with Random Forest, identifying key risk factors like excessive daily usage and notification frequency.

To further emphasize the relationship between mobile usage and addiction, Figure 8 below revisits the scatter plot of daily usage time versus addiction score, highlighting the correlation of 0.65 that underscores the importance of usage time as a predictor.

Figure 8: Daily Usage Time vs. Addiction Score (Revisited)

Inference on Daily Usage Time vs. Addiction Score Scatter Plot: When I look at this scatter plot, it really tells a story about how our phone habits might be affecting us more than we realize. Each dot represents one of the 302 students we studied, showing how many hours they spend on their phone each day and how they rated their own addiction on a scale from 1 to 10. That red line running through the dots shows a correlation of 0.65, which means there's a pretty strong link between the two—the more time you spend on your phone, the more likely you are to feel addicted to it.

What stands out to me is how some students are clocking 8 or even 10 hours a day on their phones, and their addiction scores are up at 7 or 8. That's a lot of time—like, almost half the day! I can't help but think about what that means for their studies or even just hanging out with friends in person. On the other hand, students who use their phones for just 2 or 3 hours tend to score lower, around 3 or 4, which feels like they've got a healthier balance. It's like the phone starts taking over the more you let it into your day, pulling you in deeper.

This graph really backs up why our Random Forest model flagged daily usage time as one of the top predictors of addiction—it's not just a number; it's a real pattern in how we live. Seeing this makes me feel even more strongly about the solutions we're suggesting, like setting up “no-phone” zones or teaching digital wellness in class. If we can help students cut down even a couple of hours, it might make a big difference in keeping that addiction score from creeping up. It's a reminder to all of us to maybe check our own phone timers and think twice about how much we're letting our screens run the show.

Recommendations include integrating digital wellness education into academic curricula, expanding counseling services, and implementing stricter mobile usage policies in educational settings. Institutions should establish “no-phone” zones during classes and leverage technology for early intervention, such as mobile apps for self-monitoring screen time. Future initiatives could focus on longitudinal studies to track addiction trends, develop AI-driven intervention tools, and advocate for state-level policies to address digital addiction comprehensively. By adopting these strategies, Karnataka can mitigate mobile addiction and foster a healthier academic environment for students.

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